STUDENT NUMBER:

APSC 171 - Term Test 1
Thursday Sep 25, 2025

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- This is a 2 hour test.
- No textbooks, notes, or aids other than your calculator (Casio 991 series) are allowed.
- For full marks, you must show all your work and explain how you arrived at your answers, unless explicitly told to do otherwise.
  Correct final answers with incorrect support may be given no credit.
- Wherever appropriate, **include units in your answers**.
- Keep and show at least three significant figures for decimal values.
- When drawing graphs, **add labels and scales on all axes**.
- If you need more room, there are blank pages at the end of the test. **If you use these pages, you must provide clear directions on the original page**, e.g. "Continued on page 10".
- Only answers written in this booklet will be graded; all scrap paper from the test will be discarded.

Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Total
7	7	6	5	8	7	5	5	50

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1. Differentiate the following functions with respect to their input variable.

   - You may assume that the input variables will always be within the domain of their respective functions.)
   - You do **not** need to simplify after you have taken the derivative.

(a) $f(x) = \sin(3x)\cos^2(x)$. [/2]

$f'(x) = 3\cos(3x)\cdot\cos^2(x) - \sin(3x)\cdot 2\cos(x)\sin(x)$

(b) $g(t) = \ln(4t^3 + 5)$. [/2]

$g'(t) = \frac{1}{4t^3+5} \cdot 12t^2$

(c) $h(r) = \frac{kr}{r^2 - R^2}$. k and R are constants. [/3]

$h'(r) = \frac{k(r^2 - R^2) - kr(2r)}{(r^2 - R^2)^2}$

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2. Consider the following vectors: $\vec{a} = \langle 2, -1, 3\rangle$, $\vec{b} = \langle 5, 0, -1\rangle$, and $\vec{c} = \langle -2, -2, -2\rangle$.

(a) Compute the length of $\vec{a}$. [/1]

$\|\vec{a}\| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}$

Answer: $||\vec{a}|| =$ $\sqrt{14}$

(b) Build a unit vector[1] in the same direction as the vector $(2\vec{b} + 3\vec{c})$. [/2]

$2\vec{b} + 3\vec{c} = 2\langle 5, 0, -1\rangle + 3\langle -2, -2, -2\rangle$

$= \langle 10, 0, -2\rangle + \langle -6, -6, -6\rangle$

$= \langle 4, -6, -8\rangle$.

length $= \sqrt{16 + 36 + 64}$

$= \sqrt{116}$

or $\langle 0.371, -0.557, -0.743\rangle$

Answer: $\frac{1}{\sqrt{116}}\langle 4, -6, -8\rangle$

(c) Find the angle between the vectors $\vec{a} = \langle 2, -1, 3\rangle$ and $\vec{b} = \langle 5, 0, -1\rangle$. Report in both degrees and radians. [/2]

$\vec{a} \cdot \vec{b} = 2 \cdot 5 + 3 \cdot (-1) = 7$

$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \, \|\vec{b}\|} = \frac{7}{\sqrt{14}\sqrt{26}} \cong 0.3669$

from (a)

$\theta \cong 1.195$ radians

Answer: the angle is 1.195 radians, or 68.5 degrees.

(d) For what value(s) of t will the vector $\langle 4, t, 1\rangle$ be perpendicular to $\vec{c} = \langle -2, -2, -2\rangle$? [/2]

$\langle 4, t, 1\rangle \cdot \langle -2, -2, -2\rangle$

$= -8 - 2t - 2 = -10 - 2t$.

Need $= 0$ for perpendicular $\Rightarrow -10 - 2t = 0$

$\Rightarrow 2t = -10 \Rightarrow \boxed{t = -5}$

Confirming: $\langle 4, -5, 1\rangle \cdot \langle -2, -2, -2\rangle = 0$ ✓

Answer: $t =$ -5

[1] A *unit vector* is a vector of length 1.

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3. The position of a particle is defined by the vector-valued function $\mathbf{r}(t) = \langle \ln(3t), 2 - t \rangle$.

(a) Eliminate the parameter to find a Cartesian equation for the trajectory of $\mathbf{r}(t)$. Express your final answer in the form $y = f(x)$. [/3]

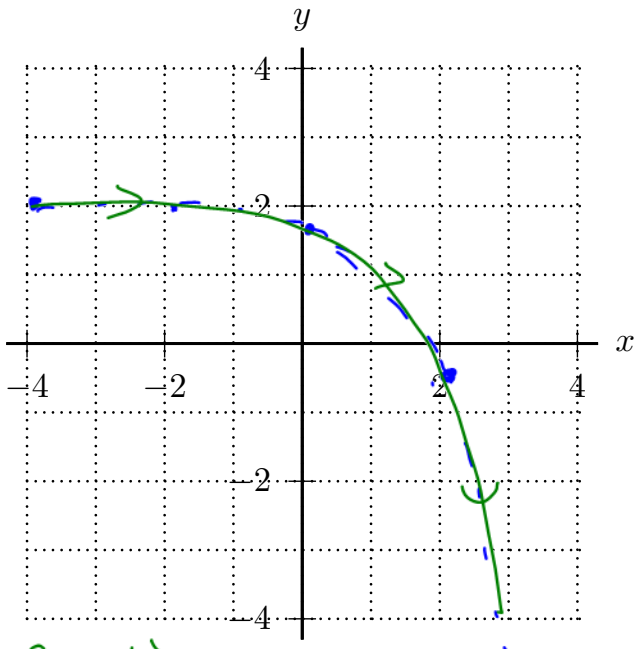
$x = \ln(3t) \rightarrow e^x = 3t \quad t = \frac{1}{3}e^x$

$y = 2 - t \rightarrow \boxed{y = 2 - \frac{1}{3}e^x}$

(b) On the axes to the right, sketch the graph of the Cartesian equation you found in part (a). Your graph should extend as far as possible on the region shown. [/3]

Once you have drawn the graph:

- Clearly indicate what part of this graph the function $\mathbf{r}(t)$ actually passes over, and
- draw arrows to indicate the direction of motion.
- Note: t can take on any real value in the domain of $\mathbf{r}(t)$.



$y = 2 - \frac{e^x}{3}$

x	y
−4	1.99
−2	1.95
0	1.67
2	−0.46
4	−16

x	y
3	−4.2

for t's $x = \ln(3t) \rightarrow$ range is all x

x increases w/ time.

$\Rightarrow$ rightward over time on the graph

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4. A trajectory is given by the vector-valued function $\mathbf{r}(t) = \left\langle t^2, \frac{1}{t} \right\rangle$, for $t > 0$.

Imagine the trajectory defines the motion of a spaceship, with a package attached to the outside. If one goal of the spaceship's mission is to have the package reach a satellite located at coordinates (12, 0), at what value(s) of t could the package be released so that it reaches the satellite? Assume after being released that no additional forces act on the package. [/5]

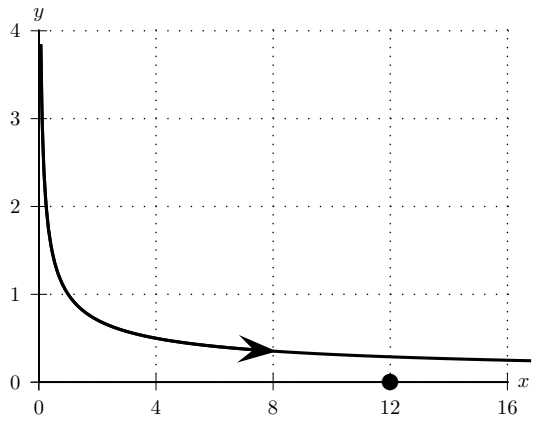
position: $\langle t^2, \frac{1}{t} \rangle$

velocity: $\langle 2t, -\frac{1}{t^2} \rangle$

If we launch at time $t=a$, the straight line trajectory will be:

$\vec{L}(t) = \langle a^2, \frac{1}{a} \rangle + \langle 2a, -\frac{1}{a^2} \rangle (t-a)$

$\vec{pos}_a$ + $\vec{v}_a \cdot (t-a)$



For this straight line to hit (12,0) we need

x: $12 = a^2 + 2a(t-a)$ ①

y: $0 = \frac{1}{a} - \frac{1}{a^2}(t-a)$ ② $\rightarrow \times a^2$

$0 = a - (t-a)$

$0 = 2a - t$

$\boxed{t = 2a} \rightarrow$ ①

$t = 2a$

$12 = a^2 + 2a(2a - a)$

$12 = a^2 + 2a^2$

$12 = 3a^2$

$4 = a^2$

$\boxed{a = 2}$, -2 ← ignore b/c we were told to use $t > 0$ for this trajectory

→ we should launch at $t=2$ to hit the target.

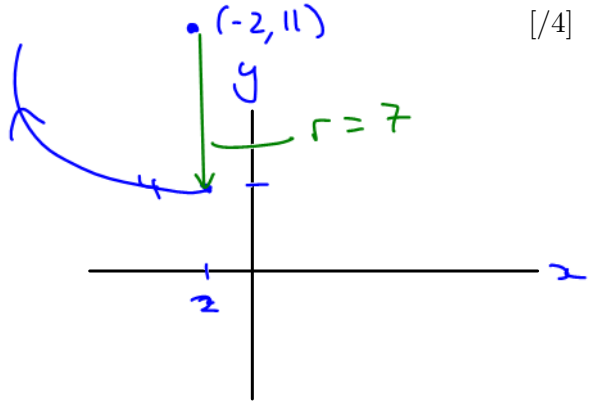
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5. (a) Define a vector-valued function for the path of a particle in $\mathbb{R}^2$ that:
   - begins at $(x, y) = (-2, 4)$ at $t = 0$;
   - moves **clockwise** around a circle of radius 7, centered at the point (-2, 11),
   - moves at a constant speed of 6 m/s. (Assume t is in seconds, and x and y are in meters.)

[/4]

$\vec{u}(t) = \langle -7\sin\left(\frac{6}{7}t\right) - 2,$

$-7\cos\left(\frac{6}{7}t\right) + 11 \rangle$



(b) Verify that your answer to part (a) satisfies the speed requirement by finding a general formula for the velocity of the particle, and then the resulting speed. [/2]

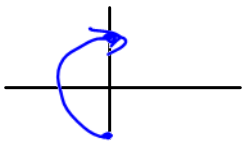
$\vec{v}(t) = \frac{d}{dt}\vec{u}(t) = \langle -7\cos\left(\frac{6}{7}t\right)\cdot\frac{6}{7},\ 7\sin\left(\frac{6}{7}t\right)\cdot\frac{6}{7} \rangle$

$= \langle -6\cos\left(\frac{6}{7}t\right),\ 6\sin\left(\frac{6}{7}t\right) \rangle$

$\text{speed} = \|\vec{v}\| = \sqrt{6^2\cos^2\left(\frac{6}{7}t\right) + 6^2\sin^2\left(\frac{6}{7}t\right)}$

$= \sqrt{36}\ \sqrt{\cos^2\left(\frac{6}{7}t\right) + \sin^2\left(\frac{6}{7}t\right)} = 6.$ ✓

(c) Find a value of t at which this particle would be moving in the same direction as the $\langle 1, 0 \rangle$ vector, i.e. moving rightward on an xy graph, with a 0 vertical component. [/2]



$\frac{1}{2}$ cycle after $t=0$.

$\text{period} = \frac{2\pi}{6/7} = \frac{7\pi}{3}$ s

$\Rightarrow$ at $t = \frac{1}{2}\left(\frac{7\pi}{3}\right) = \frac{7\pi}{6} \approx 3.665$

Checking: $\vec{v}\left(\frac{7\pi}{6}\right) = \langle -6\cos\left(\frac{6}{7}\frac{7\pi}{6}\right),\ 6\sin\left(\frac{6}{7}\frac{6\pi}{6}\right) \rangle$

$= \langle 1, 0 \rangle$ ✓

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6. The following questions are not related to each other.

(a) The trajectories defined by $s_1(t) = \langle 3e^{t+1}, t^2 \rangle$ and $s_2(t) = \langle 4+t, t^4 \rangle$ collide. Find the time(s) and the xy location(s) of any collisions. [/3]

x: $3e^{t+1} = 4+t$ ①

y: $t^2 = t^4 \rightarrow t = 1, -1$ ②

Case $t=1$ in ①: LHS $= 3e^2$, RHS $= 4+1 = 5$ $\neq$ no solution

Case $t=-1$ in ①: LHS $= 3e^0 = 3$, RHS $= 4-1 = 3$ ✓

At $t=-1$ the particles collide at $\langle 3, 1 \rangle$

(b) Consider the two trajectories in $\mathbb{R}^2$ defined by
$\mathbf{r}(t) = \langle c+t,\ c+t,\ 5+t \rangle$ and $\mathbf{u}(t) = \langle t,\ 1-t,\ 3+t \rangle$,
where c is a constant.
Find any value(s) of c for which the $\mathbf{r}(t)$ and $\mathbf{u}(t)$ trajectories would intersect. For each c value, also report the (x, y, z) coordinates where the trajectories intersect. [/4]

Using 's' for 't' in $\vec{u}$:

x: $c+t = s$ ①

y: $c+t = 1-s$ ②

z: $5+t = 3+s$ ③

① − ② $\quad s = 1-s$

$2s = 1$

$\boxed{s = \frac{1}{2}}$ ④

④ → ③ $\quad 5+t = 3+\frac{1}{2}$

⑤ $\boxed{t = -\frac{3}{2}}$

④, ⑤ → ① $\quad c + \left(-\frac{3}{2}\right) = \frac{1}{2}$

$\boxed{c = 2}$

The paths will intersect if $c=2$.

location: @ $t = -\frac{3}{2}$, $c = 2$,

$\vec{r}\left(-\frac{3}{2}\right) = \left\langle 2+\left(-\frac{3}{2}\right), 2+\left(-\frac{3}{2}\right), 5+\left(-\frac{3}{2}\right) \right\rangle$

$\boxed{= \left\langle \frac{1}{2}, \frac{1}{2}, \frac{7}{2} \right\rangle}$ check: $\vec{u}\left(\frac{1}{2}\right) = \left\langle \frac{1}{2}, \frac{1}{2}, \frac{7}{2} \right\rangle$ ✓

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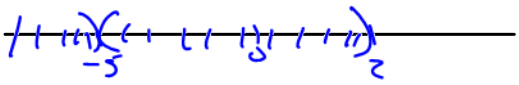
7. The following questions are not related to each other.

(a) Find the domain of the vector-valued function $\vec{u}(t) = \left\langle \ln(2-t),\ \dfrac{t}{t+5} \right\rangle$.
Express your answer using inequalities or interval notation. [/2]

$\ln(2-t)$ defined on'y for $2-t>0$

$2>t$ or $t<2$

$\frac{t}{t+5}$ defined for all $t \neq -5$



$t \in (-\infty, -5) \cup (-5, 2)$ or $t < -5$ or $-5 < t < 2$.

(b) Show that all points of the trajectory

$$\vec{r}(t) = \langle \cos(t)\sin(2t),\ \sin(t),\ \cos(t)\cos(2t) \rangle$$

lie on the unit sphere, defined by $x^2 + y^2 + z^2 = 1$. [/3]

LHS $= x^2 + y^2 + z^2$

$= \cos^2(t)\sin^2(2t) + \sin^2(t) + \cos^2(t)\cos^2(2t)$

group terms w/ $\cos^2(t)$'s $= \cos^2(t)\left(\sin^2(2t) + \cos^2(2t)\right) + \sin^2(t)$

$= \cos^2(t) + \sin^2(t)$

$= 1$

RHS $= 1$ } equal for all t's

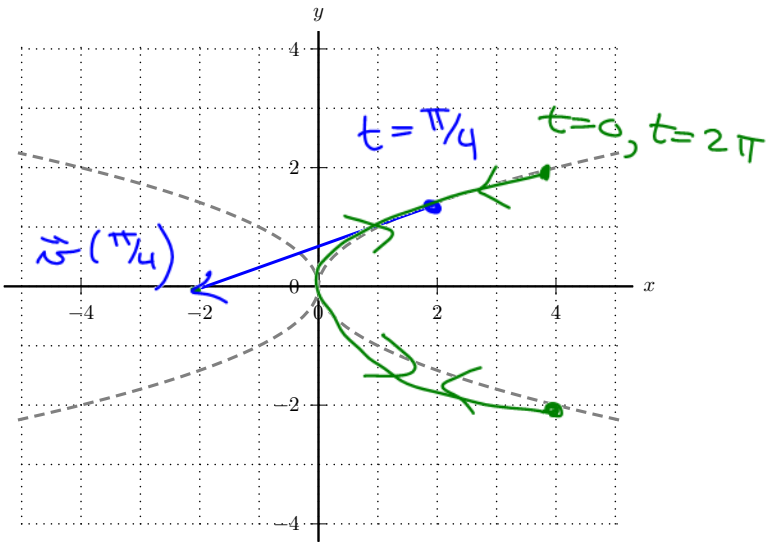
$\Rightarrow$ for every t, $\vec{u}(t)$ lies on the sphere $x^2 + y^2 + z^2 = 1$.

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8. The trajectory of a particle is given by

$$\vec{r}(t) = \langle 4\cos^2(t),\ 2\cos(t) \rangle$$

lies on part of the curve shown in the dashed line on the axes to the right.



(a) Find the position of the particle at $t = \frac{\pi}{4}$. Indicate this position on the graph. [/1]

$\vec{r}\left(\frac{\pi}{4}\right) = \left\langle 4 \cdot \left(\frac{1}{\sqrt{2}}\right)^2,\ 2\frac{1}{\sqrt{2}} \right\rangle$

$= \langle 2, \sqrt{2} \rangle \approx \langle 2, 1.41 \rangle$

(b) Find the velocity of the particle at $t = \frac{\pi}{4}$. Draw the velocity vector on the graph at an appropriate location. [/2]

$\vec{v}(t) = \langle -8\cos(t)\sin(t),\ -2\sin(t) \rangle$

$\vec{v}\left(\frac{\pi}{4}\right) = \left\langle -8\left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right),\ -2\frac{1}{\sqrt{2}} \right\rangle \approx \langle -4, -1.41 \rangle$

(c) On the graph, fill in the portion of the Cartesian graph that will be traversed by the particle over the specific time interval $0 \le t \le 2\pi$. Indicate with arrows the direction of travel between $t = 0$ and $t = 2\pi$. [/2]

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Space for additional work. **Indicate clearly which question you are continuing if you use this space.**

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